AP & RAP

Quiz 3

Date: 17 December 2025
Time allowed: 60 minutes **Full mark: 100**

1. (Definitions and examples, 25 points)
   (a) Provide the definitions of convergence in probability and in $\mathcal{L}^p$.
   (b) Provide an example of a sequence of random variables that converges in probabilty but not almost surely.
   (c) Provide an example of a sequence of random variables that converges almost surely but not in $\mathcal{L}^p$.
   (d) Provide the definiton of a Cauchy sequence in $\mathcal{L}^p$, where $p \geq 1$.
   (e) Provide the definition of uniformly p-integrable family of random variables.
2. (Short questions, 20 points) In the following questions, no justification is necessary. Simply provide the shortest possible correct response. Each question is worth 5 points.
   (a) Let X, Y be random variables that are in $\mathcal{L}^p$, where $p \geq 1$ is a constant. Suppose $\alpha, \beta \in \mathbb{R}$. Is $\alpha X + \beta Y$ also in $\mathcal{L}^p$? If yes, find a upper bound for the $\mathcal{L}^p$ norm of $\alpha X + \beta Y$.
   (b) The space $\mathcal{L}^p$ is complete for any $p > 0$. Is this statement true?
   (c) Let X be a random variable with $\mathbb{E}[|X|^p] < \infty$, where $p > 0$. Is it always true that $\mathbb{E}[|X|^q] < \infty$ if $q \in (0, p)$? Explain.
   (d) Suppose random variables X_n converges in $\mathcal{L}^p$ to X, where $p \geq 1$. Is it true that X_n converges in probability to X and $\|X_n\|_p \to \|X\|_p$ as $n \to \infty$?
3. (Theorems and proofs, 20 points) Please state the weak law of large numbers and prove it.
4. (20 points) Prove: If $X_n \to X$ converges in probability, and $Y_n \to Y$ converges in probability, then $X_n \pm Y_n \to X \pm Y$ converges in probability, and $X_n Y_n \to XY$ converges in probability.
5. (15 points) Suppose $X, Y, Y_1, Y_2 \ldots$ are random variables. Suppose $X \geq 0$, $E[X] < \infty$ and $|Y_n| \leq X$ for all $n = 1, 2, \ldots$, and Y_n converges to Y in probability. Show that Y is integrable and $E[Y_n] \to E[Y]$.